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QUESTION 3 • UNIT 2 – ACTIVATION FUNCTIONS

What is an Activation Function? Enlist and Explain different Activation functions used in Neural Network.

📝 5 Marks

Activation Functions in Neural Networks

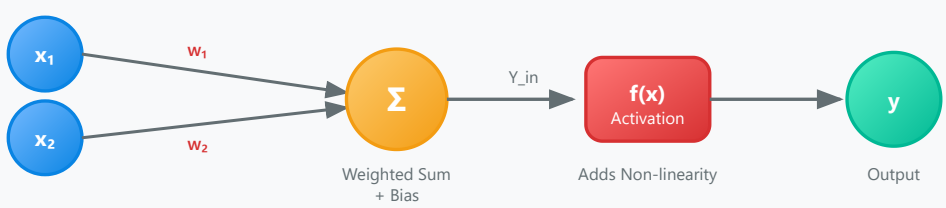
Complete answer with curve diagrams for every function

What is an Activation Function?

An activation function is a mathematical function applied to the output of a neuron in a neural network. It determines whether a neuron should be activated or not, introducing **non-linearity** into the model.

This non-linearity allows the network to learn complex patterns and relationships in data. Without activation functions, even a deep network would behave like a single linear transformation — unable to solve problems like image recognition or NLP.

Figure 1 — Role of the Activation Function in a Neuron



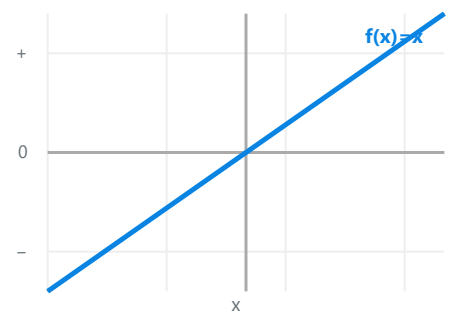
The activation function f(x) sits after the weighted summation and transforms Y_in into the neuron's output y.

Types of Activation Functions

1. Linear Activation Function

$$f(x) = x$$

Output is directly proportional to input. Rarely used in practice — does **not introduce non-linearity**, so a deep network with only linear activations collapses to a single linear layer. Cannot solve complex problems.

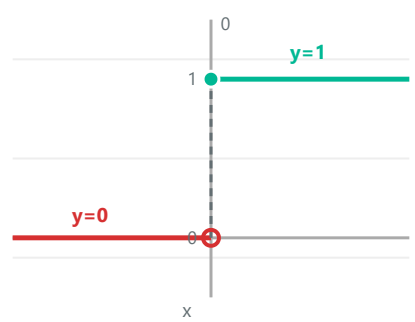


Straight line through origin

2. Binary Step Function

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

Outputs 0 or 1 based on threshold at x = 0. Used in single-layer networks for **binary classification**. Not differentiable at x = 0, limiting its use in backpropagation.

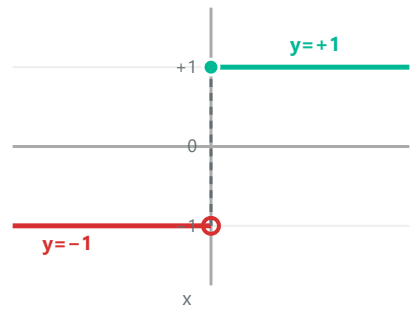


Sharp step at x = 0

3. Bipolar Step Function

$$f(x) = \begin{cases} -1 & \text{if } x < 0 \\ +1 & \text{if } x \geq 0 \end{cases}$$

Outputs −1 or 1. Similar to binary step but symmetric about the x-axis. Used for **bipolar classification** in single-layer networks.

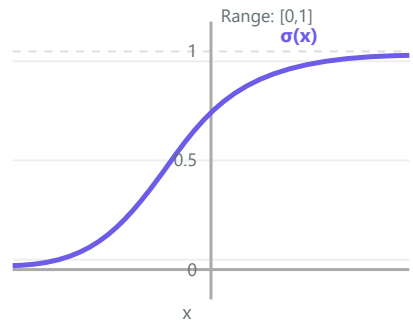


Symmetric step: −1 or +1

4. Binary Sigmoidal Function

$$f(x) = \frac{1}{1 + e^{-Ax}}$$

Smooth S-curve producing output between **0 and 1**. The steepness parameter A controls how sharp the curve is. Commonly used in **back-propagation networks** and as a probability estimator.

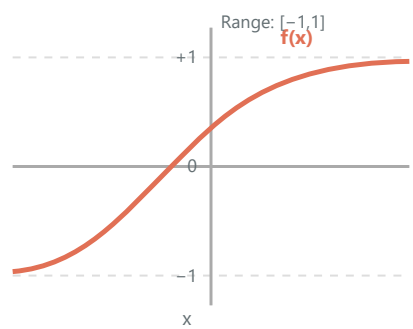


S-curve output: 0 to 1

5. Bipolar Sigmoidal Function

$$f(x) = \frac{2}{1 + e^{-Ax}} - 1$$

Smooth S-curve producing output between **−1 and 1**. It is a shifted and scaled version of the binary sigmoid. Also used in **back-propagation networks**.

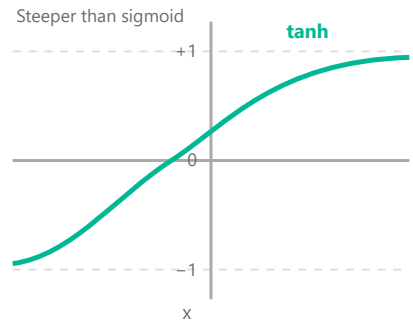


S-curve output: −1 to 1

6. Tanh Activation Function

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Output range **−1 to 1**, zero-centred. Preferred over sigmoid for multi-layer networks as zero-centring makes optimization easier. Stronger gradients than sigmoid near x = 0.

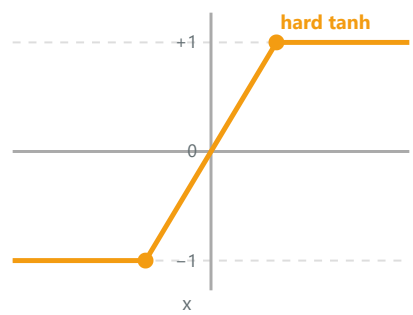


Zero-centred, range [−1,1]

7. Hard Tanh Activation Function

$$f(x) = \begin{cases} -1 & \text{if } x < -1 \\ x & \text{if } -1 \le x \le 1 \\ +1 & \text{if } x > 1 \end{cases}$$

A computationally cheaper **piecewise linear approximation** of tanh. Uses clipping — outputs −1 or +1 at extremes and identity in between.

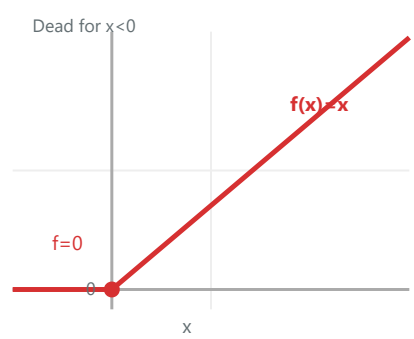


Piecewise linear clipping

8. ReLU (Rectified Linear Unit)

$$f(x) = \max(0, x)$$

Outputs 0 for all negative inputs and the raw value for positives. **Most widely used** activation in deep networks — computationally cheap, avoids the **vanishing gradient problem**, and enables fast convergence.

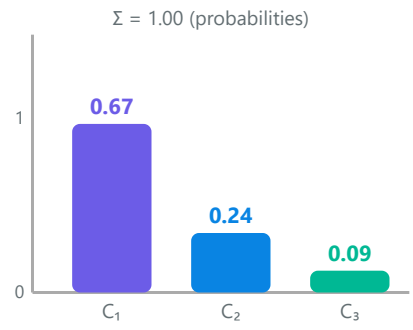


Zero for x<0, linear for x≥0

9. Softmax Activation Function

$$f(x_i) = \frac{e^{x_i}}{\Sigma_j e^{x_j}}$$

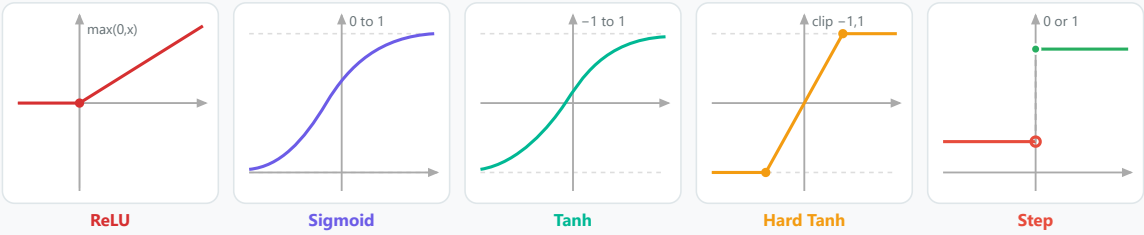
Converts a vector of real numbers into a **probability distribution** (all outputs sum to 1). Used exclusively in the **output layer for multi-class classification**. Larger inputs get proportionally higher probabilities.



Output sums to 1 across all classes

Visual Comparison — All Activation Functions at a Glance

Figure 2 — Side-by-side Comparison of Key Activation Curves



← Compare how each function transforms the same input range →

Key activation functions side by side — note how ReLU stays linear for positive values, sigmoid/tanh produce smooth S-curves, and step functions produce sharp transitions.

Importance of Activation Functions

- Introduce **non-linearity**, enabling the network to learn complex, real-world patterns.
- Determine whether a neuron should **fire or remain dormant**, controlling information flow.
- Play a crucial role in the **convergence speed** of the model during training.
- Choice of activation function directly impacts susceptibility to the **vanishing/exploding gradient problem**.